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Real-time visual intelligence for defect detection in pharmaceutical packaging

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Defect detection in pharmaceutical blister packages is the most challenging task to get an accurate result in detecting defects that arise in tablets while manufacturing. Conventional defect detection methods include human intervention to check the quality of tablets within the blister packages, which is inefficient, time-consuming, and increases labor costs. To mitigate this issue, the YOLO family is primarily used in many industries for real-time defect detection in continuous production. To enhance the feature extraction capability and reduce the computational overhead in a real-time environment, the CBS-YOLOv8 is proposed by enhancing the YOLOv8 model. In the proposed CBS-YOLOv8, coordinate attention is introduced to improve the feature extraction capability by capturing the spatial and cross-channel information and also maintaining the long-range dependencies. The BiFPN (weighted bi-directional feature pyramid network) is also introduced in YOLOv8 to enhance the feature fusion at each convolution layer to avoid more precise information loss. The model's efficiency is enhanced through the implementation of SimSPPF (simple spatial pyramid pooling fast), which reduces computational demands and model complexity, resulting in improved speed. A custom dataset containing defective tablet images is used to train the proposed model. The performance of the CBS-YOLOv8 model is then evaluated by comparing it with various other models. Experimental results on the custom dataset reveal that the CBS-YOLOv8 model achieves a mAP of 97.4% and an inference speed of 79.25 FPS, outperforming other models. The proposed model is also evaluated on SESOVERA-ST saline bottle fill level monitoring dataset achieved the mAP50 of 99.3%. This demonstrates that CBS-YOLOv8 provides an optimized inspection process, enabling prompt detection and correction of defects, thus bolstering quality assurance practices in manufacturing settings.

Keywords YOLOv8, Object detection, Computer vision, Defect detection, Coordinate attention

The pharmaceutical industry employs new techniques in manufacturing processes to develop quality products. Consequently, decreasing the risk of developing faulty products. The sector is transitioning from Pharma 3.0 to Pharma 4.0, marking a significant shift in quality management approaches¹. While Pharma 3.0 relied on semi-automated processes with some human involvement, it had limitations in defect detection due to factors like human error, inconsistent lighting, tablet size variations, and fatigue. This is a critical issue, as identifying faulty tablets before packaging is essential. Pharma 4.0 addresses these challenges by incorporating high levels of automation and data analysis, powered by artificial intelligence (AI) and the Internet of Things (IoT). AI algorithms can process sensor data in real-time, enabling faster and more accurate defect identification compared to human inspectors. Computer vision (CV) technology further enhances this capability by training AI systems to replicate human vision using live video streams. This technological advancement overcomes traditional defect detection issues, leading to improved quality, increased production efficiency, enhanced safety, reduced operational costs, and decreased reliance on manual labor². The impact of these innovations is reflected in the projected growth of the pharmaceutical packaging market, which is expected to reach \$229.9 billion by 2027, up from \$99.9 billion

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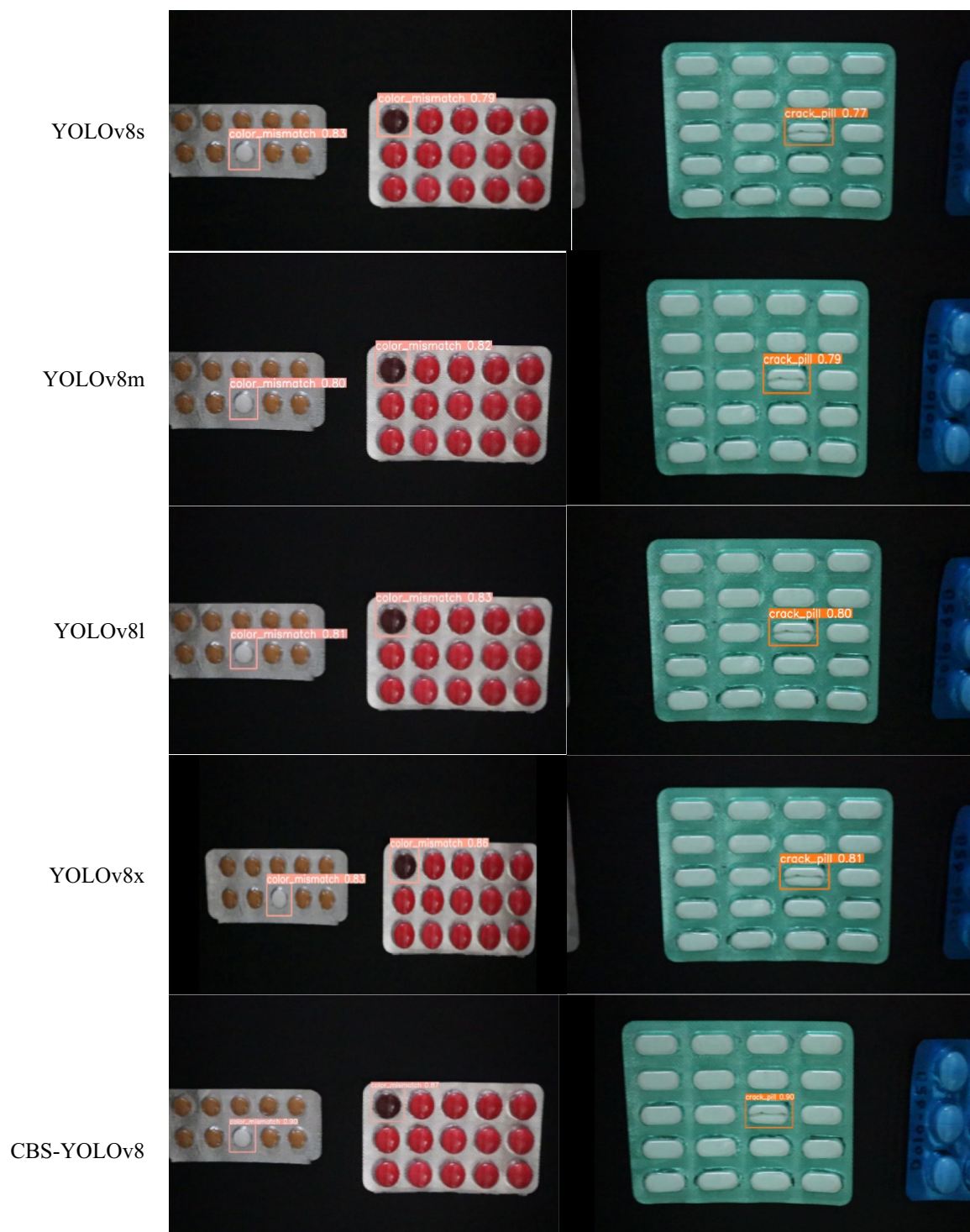


Figure 14. (continued)

Conclusion

In order to increase the feature extraction capacity in YOLOv8, this research work proposes the CBS-YOLOv8 model for detecting defects in pharmaceutical blister packages. This model incorporates the CA mechanism in the backbone of the YOLOv8 model to increase the number of target features. Additionally, includes the BiFPN module in YOLOv8's neck part to enhance the multi-scale feature fusion process to minimize the important information loss while doing upsampling and downsampling because the images are in different scales. At last, to increase the inference speed of the proposed CBS-YOLOv8 model, the SPPF presented in the backbone of YOLOv8 is changed with SimSPPF, which reduces the computational overhead. The experiments are done on the custom dataset of tablet defect detection and compared with other versions of the YOLO family. The experimental results describe that the proposed CBS-YOLOv8 attains a notably higher performance than the other models.

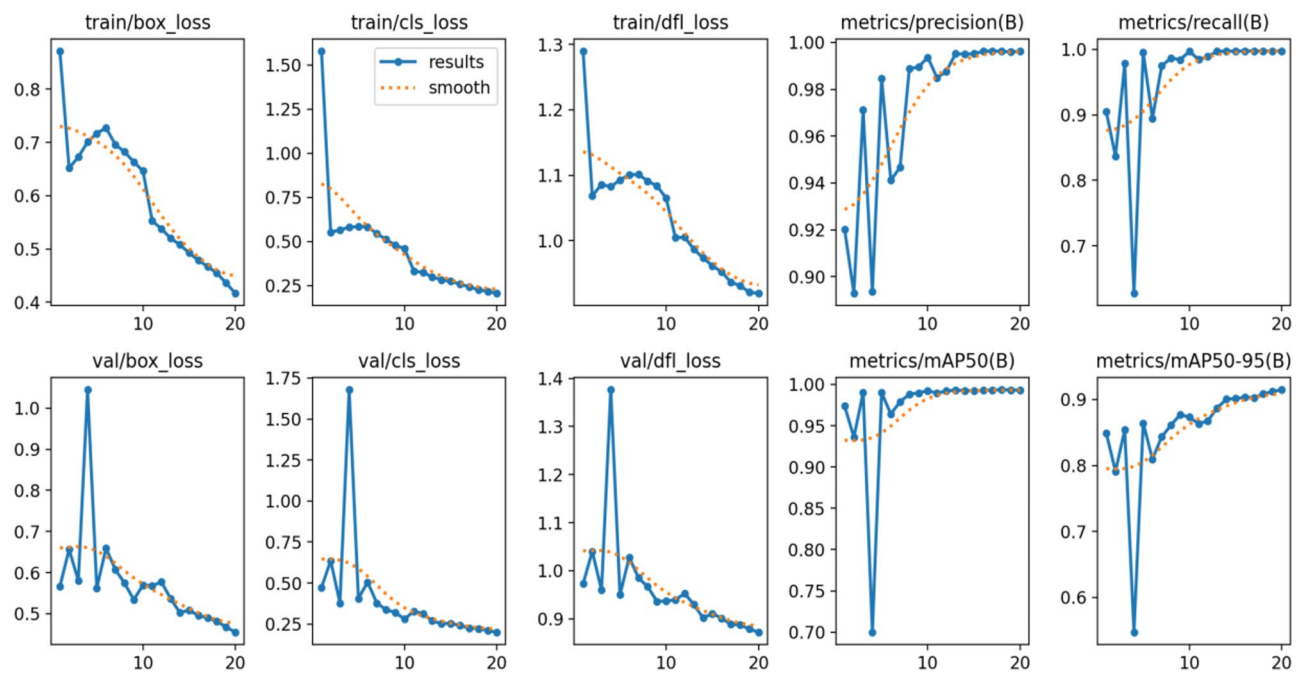


Figure 17. Loss function representation for classification loss, box loss and DFL loss for both training and validation of the CBS-YOLOv8 model on SESOVERA-ST saline bottle fill level monitoring dataset.

When equated with the baseline model YOLOv8s, the proposed model achieves a mAP50 of 97.4% on custom dataset and an inference speed of 79.25FPS, which are increased by 1.5% and 2.3%, respectively. The proposed model is also tested on the SESOVERA-ST saline bottle fill level monitoring dataset which gives mAP50 of 99.3% with high confidence value. However, the CBS-YOLOv8 achieves a better trade-off in terms of accuracy and speed. In future research, the integration of YOLO with Human-in-the-Loop (HITL) systems will be explored for managing manufacturing processes during unforeseen scenarios in which AI lacks programming to address critical situations.